

@FEUP-ASSO-2025-26 C02-T3

The Seamless Shopping Experience

Before We Begin

Problem

Traditional checkout lines create friction and diminish the in-store experience.



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Problem

Traditional checkout lines create friction and diminish the in-store experience.

The Solution

Bridge the gap between **online convenience** and **physical retail**.

Provide customers with a fully automated,
end-to-end process from GRAB-and-GO



01

System Overview

Core Technologies

Facial Recognition

Biometrical identification for secure, non-physical-contact linking of users to accounts.

RFID Tracking

Sensors provide instant detection of items as they are added or removed, ensuring a real-time digital total without manual scanning.

Smart Cart UI

A centralized interaction hub providing live feedback, GPS-guided navigation, and personalized product suggestions.



Fully Automated Journey for Registered Users

Entry

01

User passes through the gateway

Facial recognition identifies them and links their **digital account** to the **physical smart card**

Shopping

02

Items added to the cart are **detected by RFID sensors**

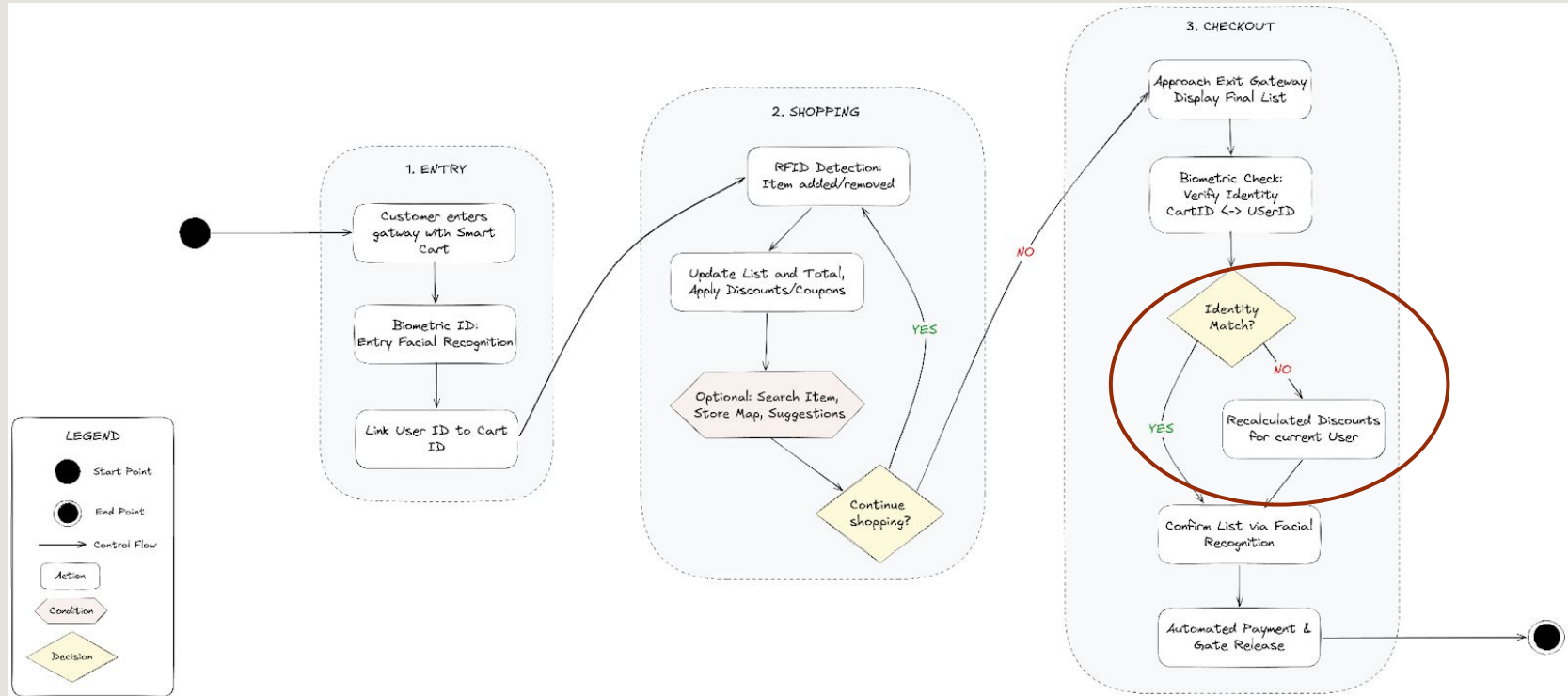
The running total and discounts sync automatically to the cart screen and phone

Checkout

03

System performs a final facial check at the exit to verify identity and **automatically processes the payment** as the gate opens

Fully Automated Journey for Registered Users



UML Activity Diagram depicting the automated "grab-and-go" journey for registered users.

02

Architectural Patterns Used

Key Architectural and Design Patterns

API Gateway

Single entry point centralizing auth, routing, and cross-cutting concerns.

Microservices

Independent deployable services for isolated scaling and fault tolerance.

Event-Driven Architecture

Async events decouple hardware from backend, enabling real-time response.

Pipes and Filters

Sequential modular steps for the pricing pipeline which is easy to extend and test.

Model-View-Controller

Async events decouple hardware from backend, enabling real-time response.

And other patterns...

03

Main Quality Attributes

Main Quality Attributes

01

Performance

Goal: Real-time responsiveness.

Requirement:

Instant updates for item detection, discount calculation, and cart totals across all screens.

**Event-Driven,
Pipes & Filters**

02

Availability

Goal: Fault tolerance.

Requirement:

Core checkout must remain active even if non-critical systems (like recommendations) fail.

Microservices

03

Security

Goal: Data protection & loss prevention.

Requirement:

Secure payments, verify identity at exit, and trigger alarms for unconfirmed walk-outs.

API Gateway

04

Modifiability

Goal: Extensibility.

Requirement:

Allow seamless updates to pricing rules and payment methods without core system overhauls or downtime.

**Pipes & Filters,
Microservices**

05

Usability

Goal: Seamless interaction.

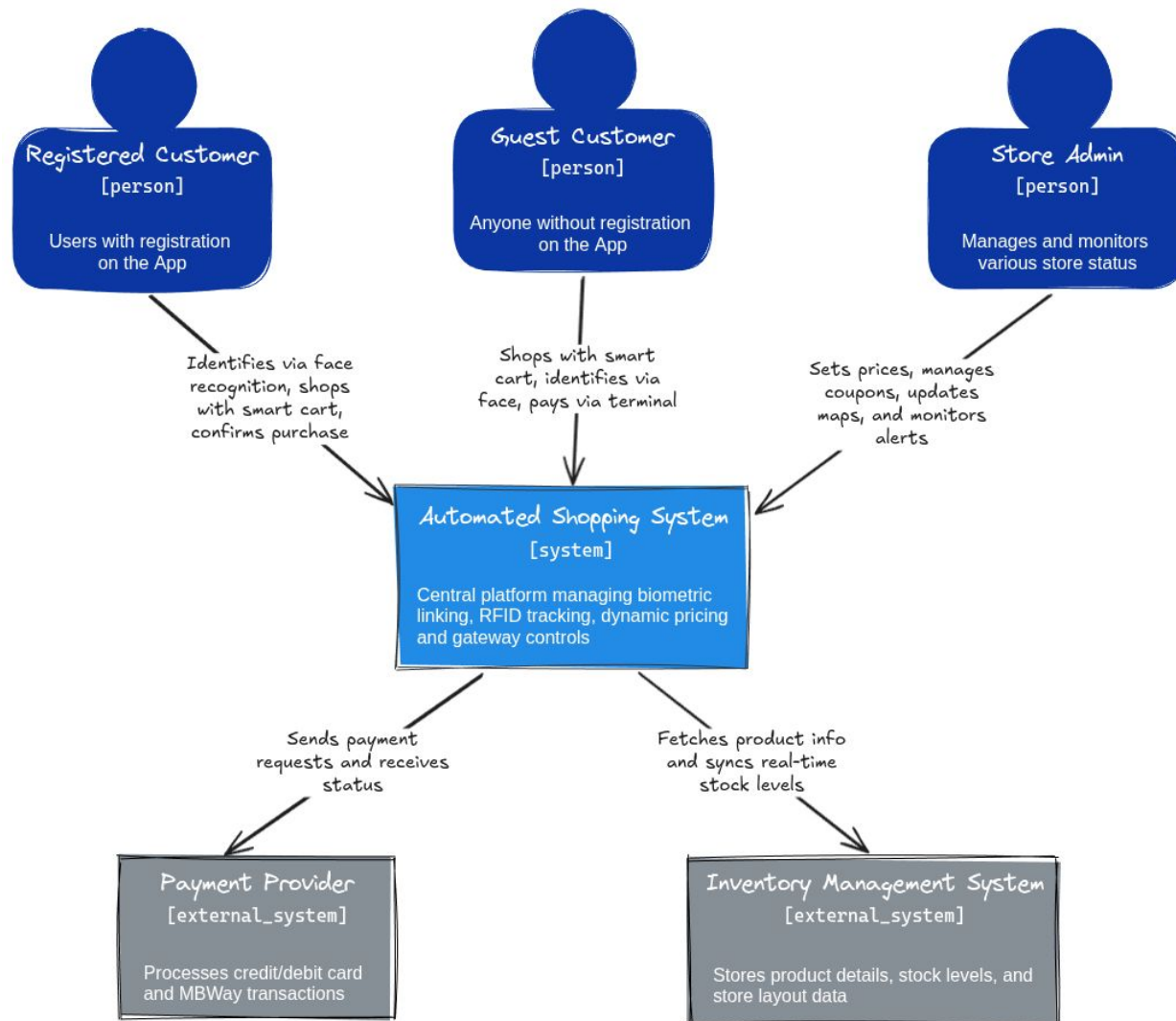
Requirement:

Ensure a consistent, intuitive experience across both mobile apps and guest cart-screens.

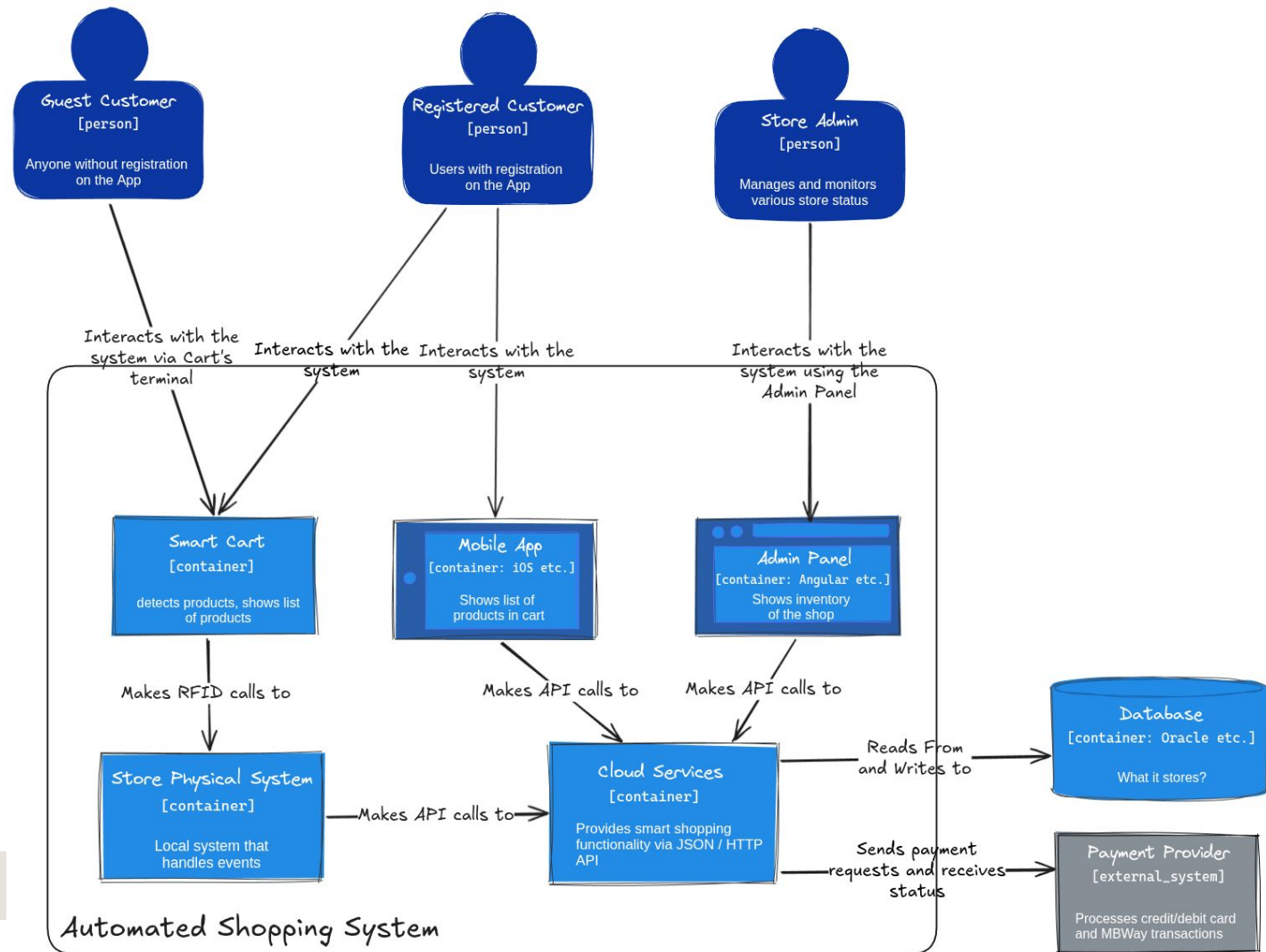
**Model-View-Cont
roller (MVC)**

04

C4 *Model* Diagrams

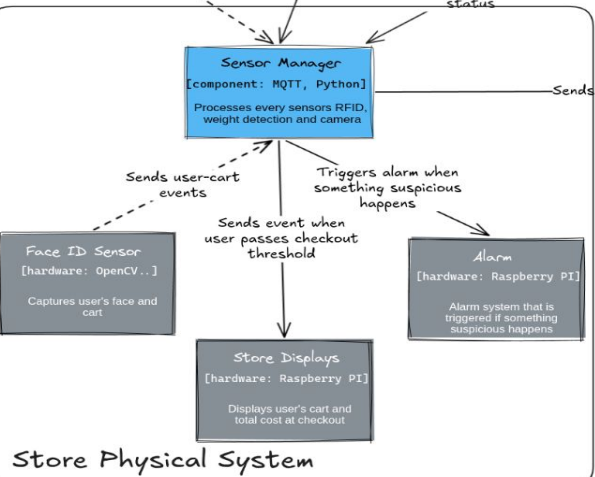
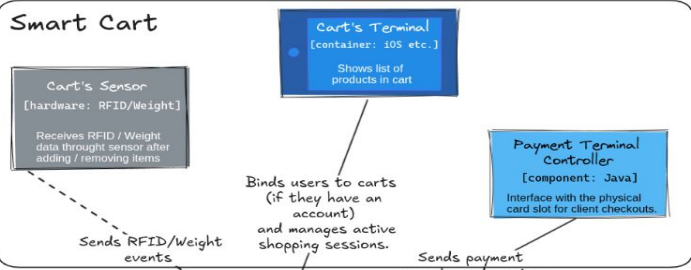


C4 Level 1 Diagram



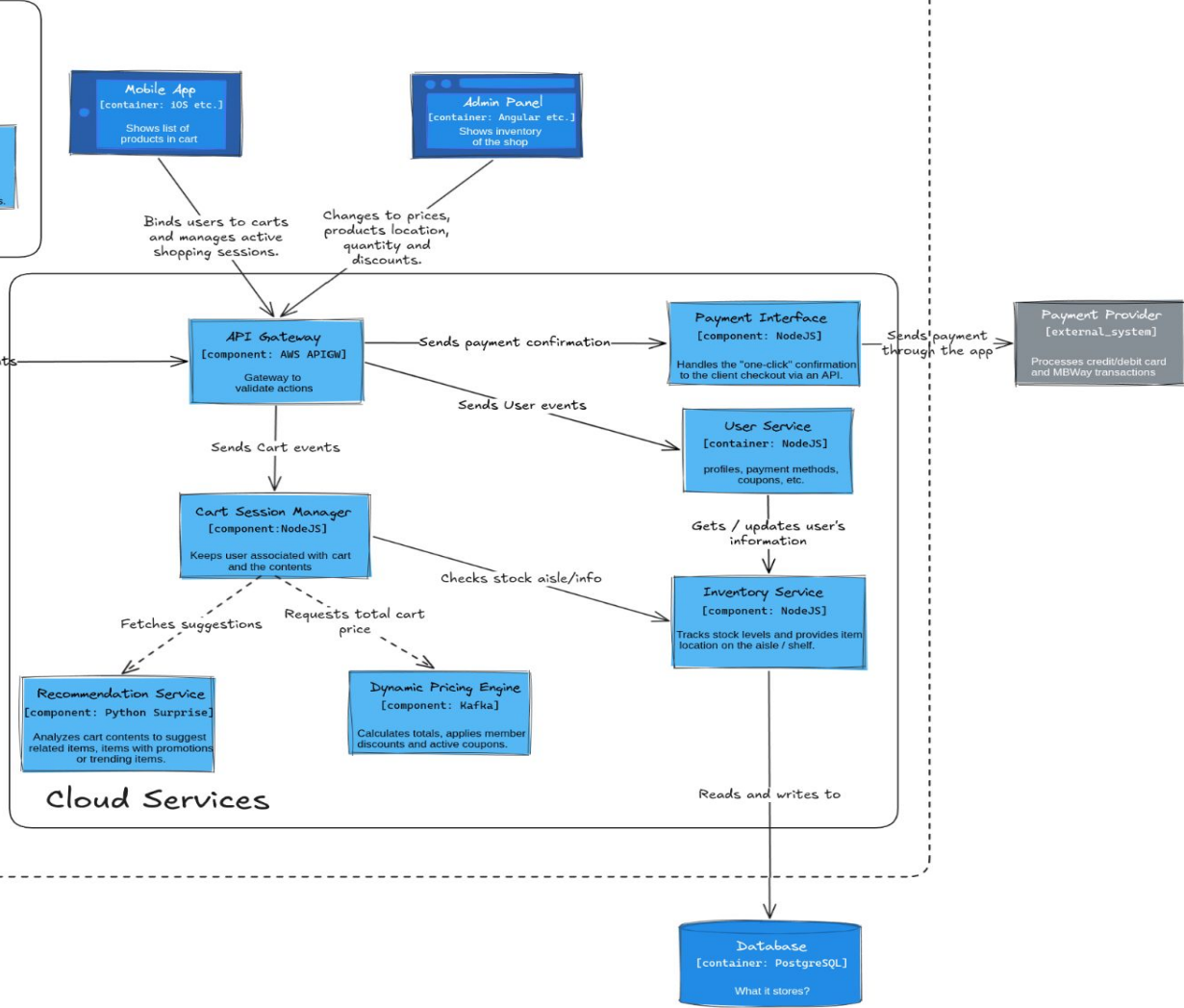
C4 Level 2 Diagram

Smart Cart



Store Physical System

Automated Shopping System



Cloud Services

05

*C*onclusion

Thank *You*

Designing the Future of Unemployed Cashiers